

Soil Water Distribution and Movement in Layered Soils of a Dam Farmland

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Abstract In soil profiles, special emphasis has been placed on the migration of agricultural chemicals spread intentionally or accidentally into deep soils or groundwater body. To prevent soil water pollution and estimate the magnitude of the hazard caused by these chemicals, it is necessary to know the processes controlling their movement from the soil surface, through the root zone and eventually to the water table. This paper deals with two-dimensional soil water distribution and movement in sloping layered soils of a dam farmland on the Loess Plateau of China. In the dam farmland, soil water content showed horizontal distribution corresponding to spatial patterns of the particle sizes. The soil water content of deeper soil was relatively stable compared with topsoil. Generally, rainfall infiltration was limited to 0.8 m in the study period. Funnel flows were found in the layered soils of the dam farmland after rainfall proving the existence of this phenomenon which was observed in simulation experiments and field observation by previous researches. In the study area, the wetting front was unstable due to the layered soils. The spatial correlation analysis of the soil water content showed water movement along the layers in the wetting process with 7 m day^{-1} only on the first day after rainfall. On the vertical direction, the velocity of water movement was 0.3 m day^{-1} on the first day after

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rainfall. The results indicated that the quantity of funnel flow increases with distance along the inclined interfaces in the dam farmland which can cause contamination of groundwater. Consequently, future studies should consider the funnel flow and the management of agriculture chemicals in dam farmlands.

Keywords Dam farmland · Layered soil · Loess Plateau · Soil water content · Wetting front

1 Introduction

In nature, the appearance of stratified soil rather than uniform soil is more common. The soil stratification affects the uniform distribution of irrigation and drainage water and the shape of soil water layer. The distribution and movement of soil water in layered soils characterised by ‘downward movement’ of water have gained more attention due to their effects on water infiltration, root growth, transportation of chemicals and pollutants, hydrological process, and ecological quality and stability (Samani et al. 1989; Antonopoulos 1993). Soil water movement and distribution studies involving layered media are important for investigating how restricting layers affect rates of water and solute migration in the soil profile and, more generally, for examining the influence of soil heterogeneity on water and solute transport.

As layered soil is a natural phenomenon that occurs in numerous soil types, various types of heterogeneity in soils associated with water distribution and movement have been observed. Miyazaki (1988) measured the water flow in unsaturated and layered soils sandwiching coarse materials in inclined soil containers. The obtained data of infiltration could be well explained by the proposed new concept of saturated and unsaturated refraction flow. The ratio of hydraulic conductivities at which the wetting front loses its one-dimensionality and breaks into “fingers” by laboratory experiments was determined by Samani et al. (1989). Finally, the impact of layered soil is important in more general studies of the effect of soil heterogeneity and anisotropy on water movement. Sakellariou-Makrantonaki (1997) compared experimental results during the first and second drainage in a vertical column of saturated layered soils to those data predicted from simulation. Gish et al. (2003) well documented the funnel flow, which describing an accelerated transport mechanism, observed under a potato field in central Wisconsin. Those results showed that more field observations under different environmental conditions were necessary to test the laboratory and field results.

Layered soils most commonly occur in fluvial deposits (Kung 1990; Boll et al. 1996). The soil of dam farmland is the fluvial deposits intercepted by the check-dams in the process of sedimentation. On the Loess Plateau of China, many dam reservoirs have been filled up with sediments within one or two decades because of serious soil erosion (Gao and Zhang 2007). The layered soils were discovered existing in the dam farmland using Cesium-137 fingerprinting technique by Zhang et al. (2007) and corresponding relationship with soil water distribution by Zhao and Shao (2010). There were 3.340 km² of dam farmlands with high crop yield 1.67 billion kg maize (*Zea mays* L.) each year (Xu et al. 2004). Dam farmland is the most important crop resource for the local farmers yielding 5–6 times the sloping farmland, even exceeding 10 times in some areas in the plateau (Xu and Wang

2000). Well soil water condition is one of the important factors contributing to high crop yield. As mentioned above, soil stratification has impacts on the water distribution and movement greatly and layered soils were found existing in dam farmland. Consequently, an understanding of soil water distribution and movement in especially eroded soil of the dam farmland is very important for the soil and water resource management, hence, crop yield. However, few studies focused on the soil water in dam farmland where the heterogeneous layered soil existed on the Loess Plateau of China as far as we know. Moreover, the studies focusing on water distribution and movement in layered soils were mainly conducted in laboratory (Stauffer and Dracos 1986; Miyazaki 1988; Samani et al. 1989; Antonopoulos 1993; Sakellariou-Makrantonaki 1997) and very few studies have conducted in field for complex field condition under tougher operations (Heilig et al. 2003; Gish et al. 2003).

As a consequence, the objectives of this study were (1) to provide observational data and describe the characteristics of the layered soils in the dam farmland; (2) to show two-dimensional soil water distribution and movement in the dam farmland; (3) to explore the existence of funnel and/or fingering flows in the dam farmland, where both flow types were observed in laboratory experiments.

2 Methods

2.1 The Study Area

The study area on the Loess Plateau and the general view of the study dam farmland is shown in Fig. 1. The soil condition of the selected dam farmland is under natural deposition without disturbance of the human activity. The experiment was carried out in the dam farmland in Liudaogou catchment ($110^{\circ}21' - 110^{\circ}23' \text{ E}$, $38^{\circ}46' - 38^{\circ}51' \text{ N}$) located at the Northwest of Shaanxi province in China (Fig. 1a). More details about this catchment can be found from the paper of Zhao et al. (2010).

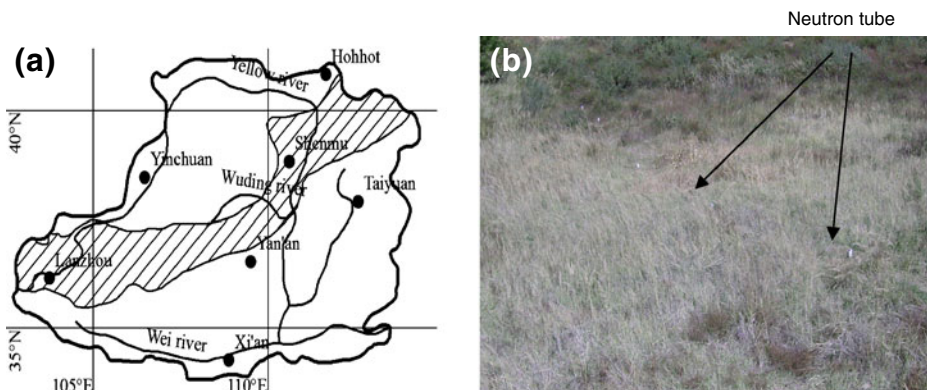


Fig. 1 The study area on the Loess Plateau (Shenmu) and the general view of the study dam farmland

2.2 Soil Moisture Observation

2.2.1 The Observation Arrangement

As shown in Fig. 2a and b, 18 observation positions were set up in the slope. There were totally 18 neutron access tubes spacing 3.5 m in the dam farmland (Fig. 2a, b). Totally 621 samples were collected with interval 10 cm depth. Those samples were offered for particle size distribution (PSD) analysis using Mastersizer 2000 (Malvern Instruments, Malvern, England). Soil type classification according to USDA, was carried out using the PSD information. There are four types of soil for all the samples: silt loam, sandy loam, loamy sand, and sand at the site (Fig. 3). After the

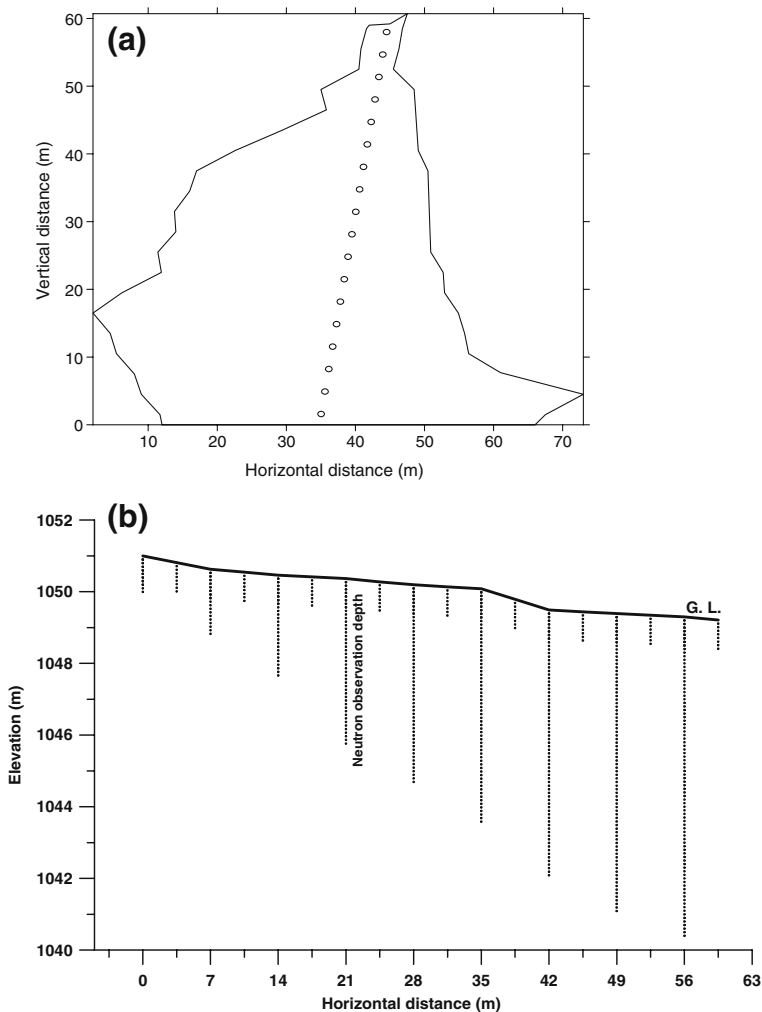


Fig. 2 The outline of sampling points on the surface and profile of the dam farmland (GL, ground level)

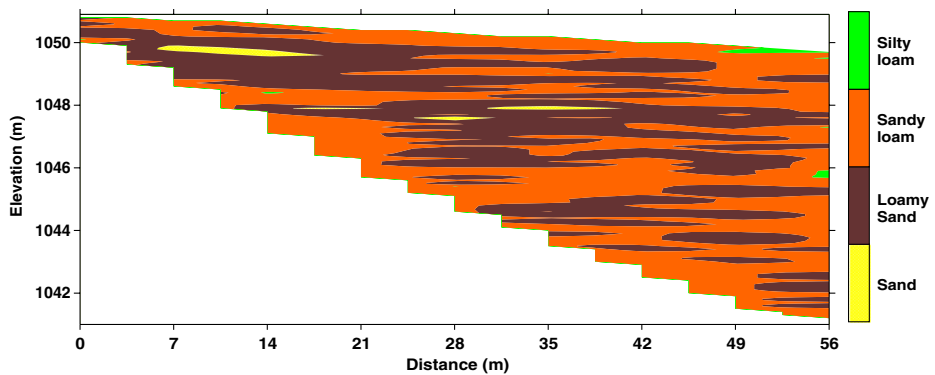


Fig. 3 The soil texture distribution at the dam farmland (USDA)

sampling, neutron tubes were carefully installed to the sampled holes in order to set the observation depth exactly at the sampled depth. 6 m was the longest tube here for it was restricted by the producer and the shallowest one was 0.8 m which was installed between two longer ones as shown in Fig. 2b. This observation style was aiming to get more soil water information with the available materials. However, 6 m long can not reach the bottom on point 13, 15, and 17 (in Fig. 2b from left to right) which the depths were 7.4, 8.3, and 8.8 m, respectively. Consequently, PSD covers more soil texture information than soil water content. The shortest tube length was decided by the result that the soil water infiltration was mainly restricted at 0.7 m in this region (She 2009). The elevation of each point was measured by GPS Trimble.

2.2.2 Neutron Tube Observation

We monitored the neutron tube every 10 days from June to September in 2008. In addition, intensive measurements for short tubes were conducted before and after rainfall. Several intensive measurements were conducted to avoid unpredictability of the time and amount of rainfall. However, only comparatively high rainfall was analyzed for the significant effect on soil water distribution and movement. The intense rainfall of 28.7 mm measurements on 7th and 8th of August were shown and analyzed in this study. The intense measurements began on the 4th of August and stopped by another rain on the 12th of August with 4.8 mm. A calculation on the water budget shows that a mean amount of 20.7 mm water infiltration into the soil profile. That well demonstrates the capture of the main infiltration pattern in the dam farmland.

2.2.3 Calibration of the Neutron Probe

For the heterogeneity of soil in the dam farmland, careful calibration work was conducted. The calibration data was classified due to PSD (USDA). There were 12 calibration data in this study located in the catchment according to the soil texture. The parameters including soil water content (dry method), bulk density, PSD, and neutron indication were used to calibrate the neutron tube. The linear fitting between neutron indication and soil water content multiplying bulk density was conducted among different soil textures. Figure 4 shows the calibration data according to the

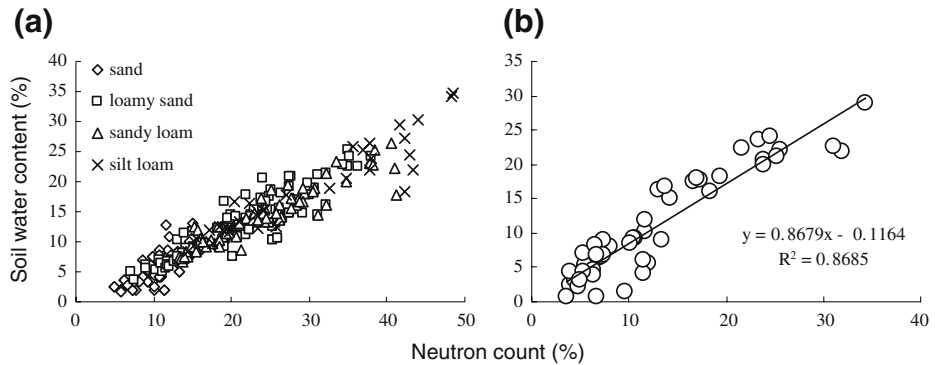


Fig. 4 The calibrations lines of the neutron tubes (**a** below 10 cm, **b** 10 cm)

soil texture. We measured soil water contents and neutron count for five times during the observation period in order to cover the lowest and the highest water content. For the calibration of topsoil water content was different to deeper layers (down to 0.2 m, Fig. 4a), we gave another calibration line for the topsoil in this study without considering of the soil texture for the limited sample number (Fig. 4b).

3 Results and Discussion

3.1 Soil Water Distribution in the Dam Farmland

The soil water distribution in the soil profile of the dam farmland from June to September is shown in Fig. 5. Figure 6 illustrates the distribution of 0.002–0.005 mm and 0.005–0.01 mm particles in the dam farmland. Obviously, the corresponding relation could be found between the soil water content and those particle size distributions, especially for the deep soil layers. It is obviously observed that the soil water content of topsoil were unstable as they were directly and significantly affected by the environmental conditions. Higher water contents were often observed in topsoil corresponding to the rainfall. Comparatively, the zone with lower soil water content was found in the middle part of the dam farmland. This could be explained by that low content of fine particles accumulate at this location (Fig. 6). In addition, the complement of soil water cannot reach that part easily for the effect of topsoil which is a barrier for the subsoil and deep soils. Hu et al. (2008) also reported that the fine particles (<0.01 mm) is related to high water content. The low water content makes some horizontal distributions of soil water disappear or be vague. Generally, the soil water contents were stable in the deep soil layers compared with topsoil. As a consequence, the soil water mainly changes significantly in the topsoil or the shallow soil layers.

3.2 Soil Water Movement in Shallow Layer Down to 0.8 m

The soil water distribution and movement in shallow layer down to 0.8 m are shown in Fig. 7. Generally, increase of soil water following rainfall was only limited in

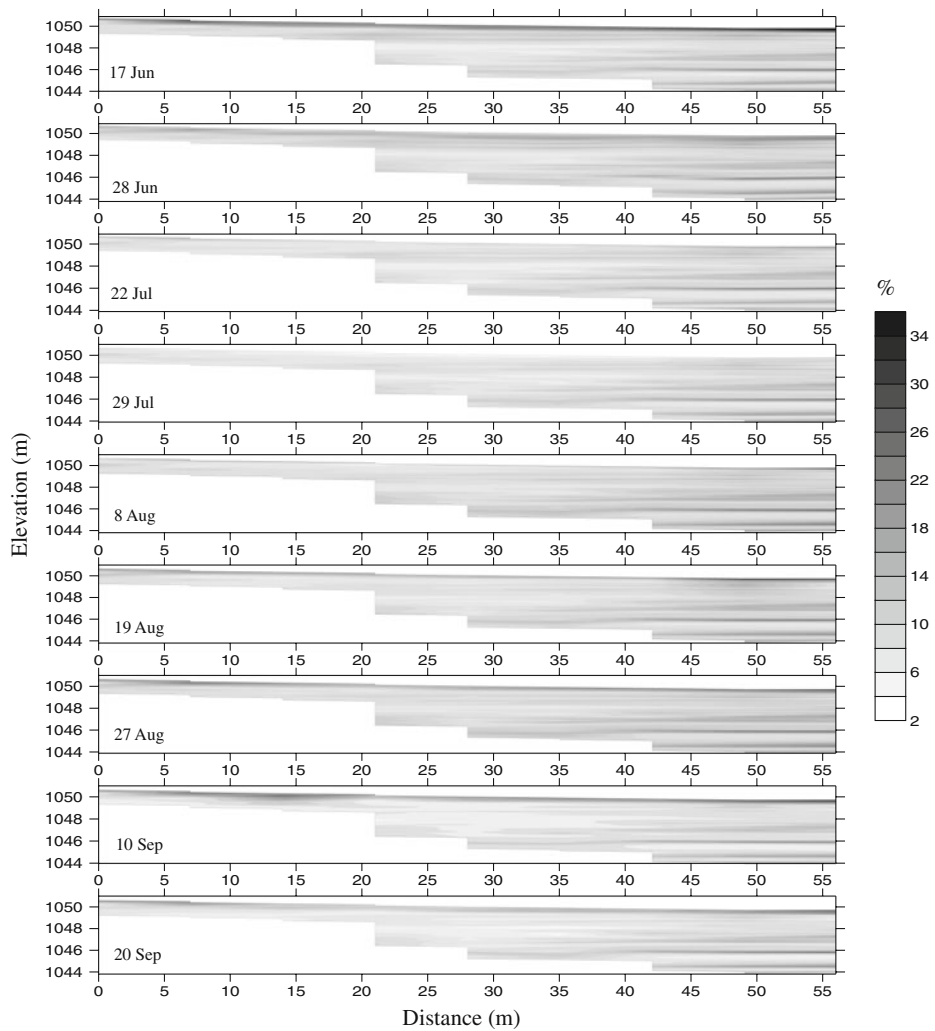


Fig. 5 The soil water distribution in the dam farmland

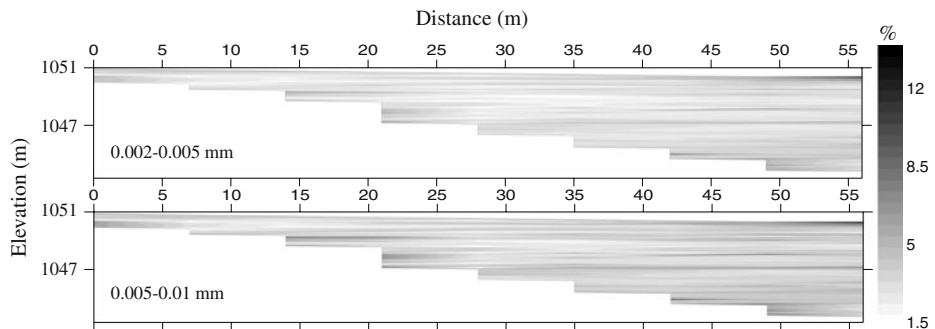


Fig. 6 The particle-sizes distributions of 0.002–0.005 mm and 0.005–0.001 mm in the dam farmland

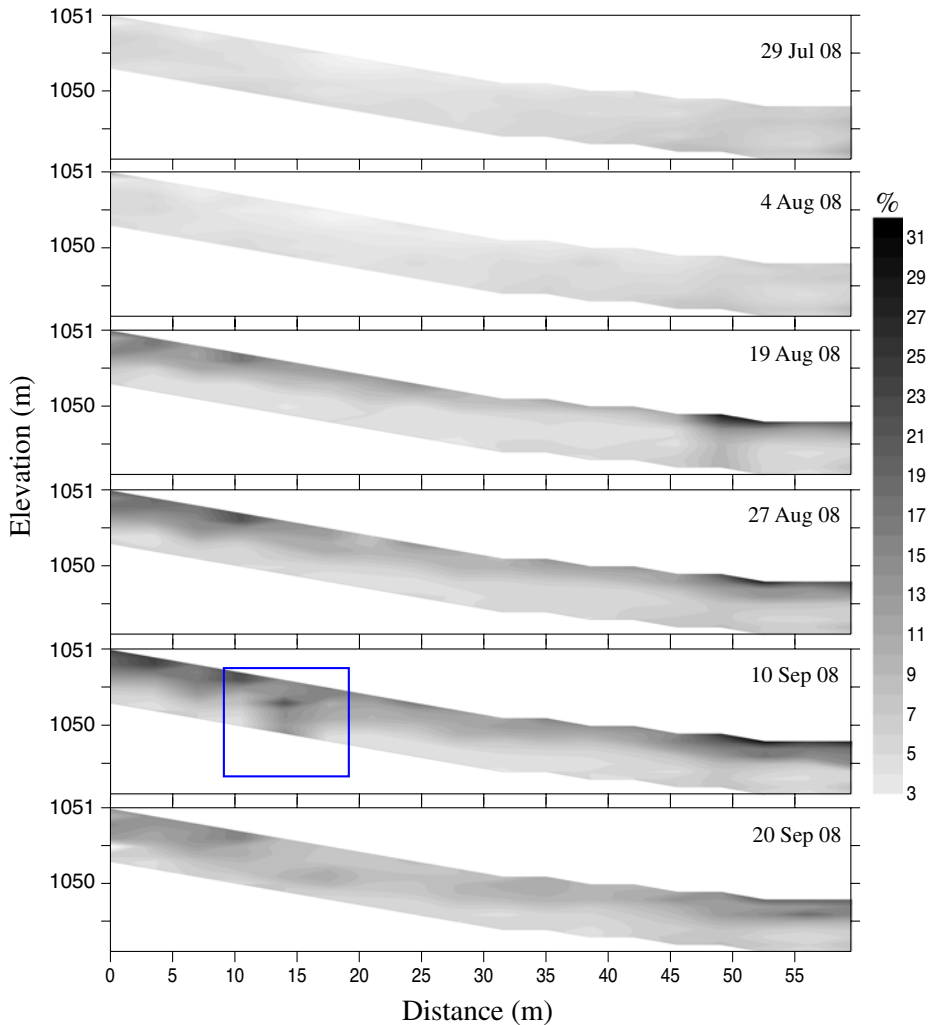


Fig. 7 Soil water distribution in the shallow layers of 80 cm depth (the square locating the funneled flow)

shallow layers (down to 0.8 m) in the investigated area. The sublayer of the middle slope was the zone with low soil water content. The toe of the dam farmland had higher water content due to the accumulation of fine particles (Figs. 3 and 6). The infiltration depth of rainfall was deeper in the upslope of the dam farmland than other locations. This can be explained by that the high proportion of sand fraction deposited in upslope of the dam farmland quickly in the sediment deposition process. Oppositely, the shallowest infiltration depth by rainfall was found in the downslope, especially at the location neighbouring the dam body where the farthest place to silting source. This phenomenon was also strongly related to PSD information with fine particles was always transported farther than coarse particles (Young 1980). Fine particles create a capillary barrier effect to limit percolation. Such an effect

was created under unsaturated conditions when a fine texture soil was placed over a coarse material. For the semi-arid climate as in the Loess Plateau, if there was no flood happening, the soil water is always under unsaturated conditions as seen in the dam farmland. From Fig. 7, the zone of low water content in sublayer was disappeared abruptly due to a rainfall infiltration on 8th of August. As many rainfalls in August, the soil water contents in the topsoil and subsurface were relatively high. Due to the accumulation of water content, the decreasing the barrier of suction made the wetting front in the dam farmland deeper than before.

3.3 Funnel Flow Observations

The funnel flow is observed in the upslope after the rainfall event (Fig. 7). This phenomenon was very obvious on the 27th of August and the 10th of September due to abundant rainfall reduces the capillary barrier efficiency of topsoil. Funnel flow is a unique category of flow phenomena referring to the situation in which a capillary barrier develops above a coarse layer which underlies a relatively fine soil (Miyazaki 1988; Kung 1990). From Fig. 3, the soil located in the place where funnel flow was observed was found with fine texture soil (sandy loam) overlaying coarse texture soil (loamy sand) which covering on a sand layer. A very thin sandy loam layer sandwiches between loamy sand soils far to 14 m to the silting source. As a consequence, the funnel flow was observed in layered soil with sandy loam overlaying loamy sand which covering on sand soil in the dam farmland. The layered soils were a three-layer soil structure existed in the dam farmland. Notice that another funnel flow is shown on the 8th of August in Fig. 7 (It is clearer in Fig. 8). The field observations of funnel flow in the dam farmland proved the existence of this phenomenon which was reported by prior researchers (Miyazaki 1988; Kung 1990; Walter et al. 2000; Heilig et al. 2003; Gish et al. 2003), but not “fingers” reported by Hill and Parlange (1972), Samani et al. (1989), and Wang et al. (2003). This can be explained that the “fingers” are very small with the diameter of only several centimeters, about 4 cm by Hill and Parlange (1972), about 6.8 ± 2.2 cm in mean size by Wang et al. (2003). They could not be easily observed in the dam farmland for the relative large scale. Generally, the instability of wetting front in the slope of layered soils of the dam farmland was observed. That meant that percolating water would use this funnel flow as avenues to recharge the water table with smaller volumes of water. On the Loess Plateau of China, dam farmland was the most important crop yield resource for local farmers. The insecticide, herbicides, and fertilizers were abundantly applied in the farmland to get high crop yield. Percolating water would use this funnel flow as thruways to recharge the water table with volumes of water than would have the wetting front remained stable. Knowledge of the development of wetting front instability is of importance to predict more closely the amount of rainfall or irrigation necessary to affect water table recharge. It also affords a possible explanation of deep penetration of mobile soil pollutants, occasionally observed in areas of limited rainfall.

3.4 Wetting Process

Figure 8 illustrates the wetting process of soil water in shallow soil layers in the dam farmland. For the wetting process, the zone of lower soil water content on

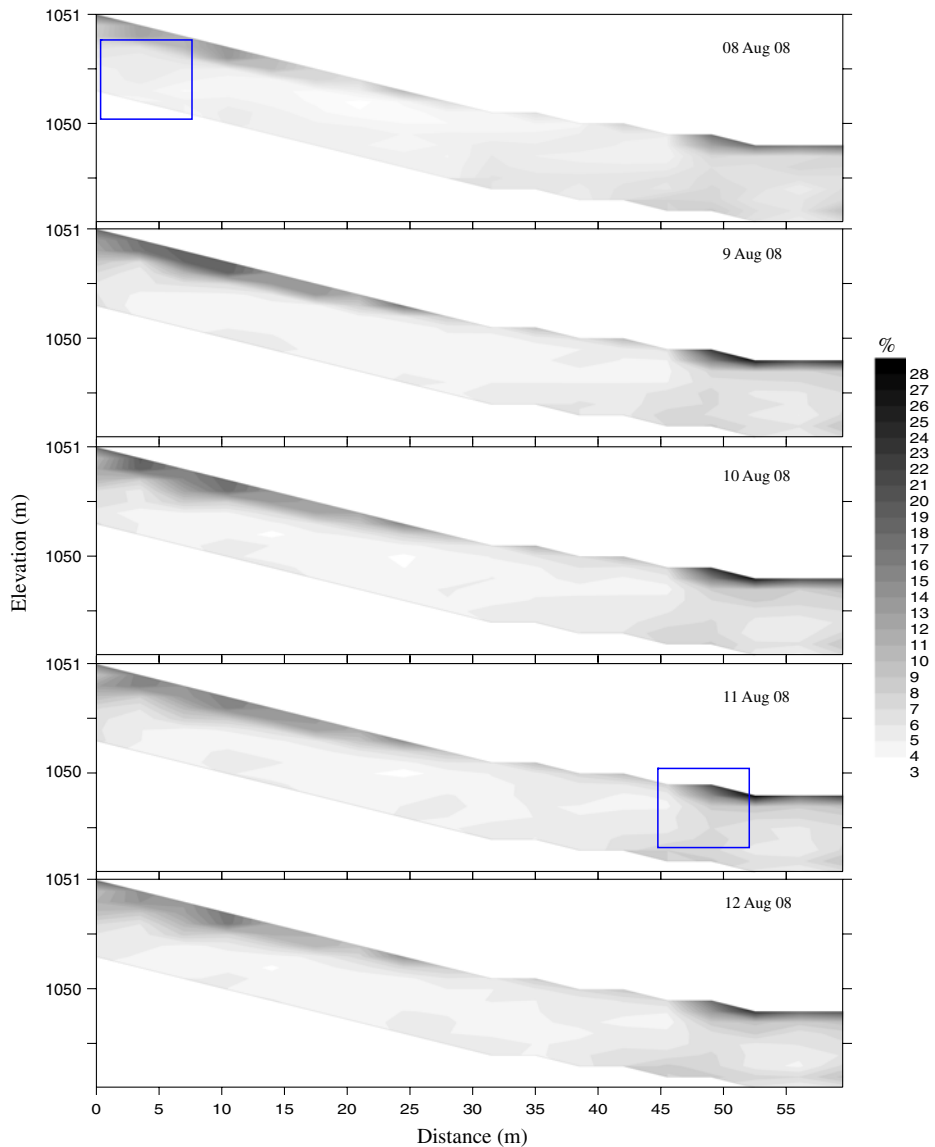


Fig. 8 The wetting process of shallow layers in the dam farmland (the square locating the funneled flow)

the 6th of August was disappeared with the rainfall infiltration. However, even the observing time of three days, no significant change was found in the subsurface of the dam farmland compared to the first day after the rainfall event. Consequently, the redistribution process of soil water in the dam farmland was slowly due to the limited water amount within a single rainfall event. This may be an explanation for the soil water infiltration limited to 0.8 m in this study. The coarse material acts as a barrier against deep percolation because of its low hydraulic conductivity at

a small degree of saturation. Water was accumulated above the interface between the two soils until the local negative pressure reaches the water entry value (WEV) of the coarse material. As the wetting process came from one rainfall event, the water amount was limited. As a consequence, the redistribution of soil water was very slow except the first day after rainfall. Hillel (1980) also stated that water at the wetting front was normally under suction, and this suction might be too large to permit entry into the relatively large pores of the coarse layer. For the structure with silt loam overlaying sandy loam, the obstruction effect on soil water infiltration and redistribution are obviously shown in Fig. 8. The rainfall had the shallowest effect depth at this location compared with others. Moreover, this kind of layered soil not only restricted the water flow downward, but also limited the lateral flow. This was proved by the phenomena of the appearance of funnel flow (downslope) connecting two relative wetting zones, and the stable water content below the layered soil (Fig. 8). This meant that the soil with fine texture located near the dam body may limit the lateral drainage in wetting periods. Generally, the soil water movement perpendicular to the layered soil could prove that the soil water movement in the dam farmland mainly depended on the suction of soil at unsaturated condition. To further clarify the observed spatial transport properties and see a tendency of soil water movement in the slope section, the spatial correlation analysis was conducted in this study. The calculation procedure could be found in the paper of Berndtsson and Chen (1994). The soil water autocorrelation field γ_φ was calculated as

$$\gamma_\varphi(\xi, \tau) = \text{corr}[\varphi(z, t), \varphi(z + \xi, t + \tau)]$$

where t is time, z is a space vector, ξ is the distance between observation points, and τ is the time lag. Figure 9 shows the correlation analysis of the wetting process in Fig. 8 of the study dam farmland. The diamond shape of correlation surfaces indicate water

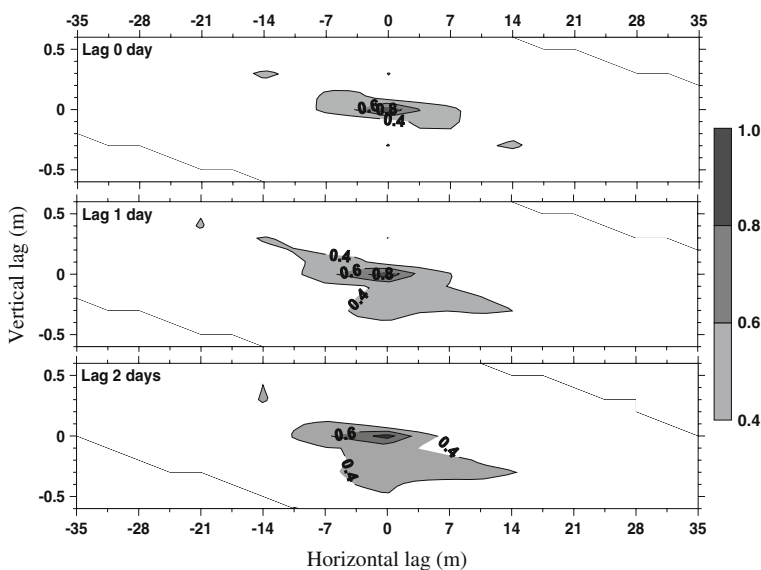


Fig. 9 Spatial correlation analysis for wetting processes in the dam farmland in shallow soil layers

movement along the layers was very pronounced from the upslope to downslope for increasing values of τ . Judging from the Fig. 9, the water layered movement velocity appeared to be about 7 m day^{-1} after rainfall. It almost not moves on the second and left days. This velocity was quicker than 3 m day^{-1} which was reported by Berndtsson et al. (1996) who study the soil water movement in a desert dune. The difference could relate to the rainfall amount, soil texture, and the slope and so on. On the vertical direction, the velocity of the water movement reaches 0.3 m day^{-1} on the first day and 0.1 m day^{-1} on 2 day lags after rainfall. Obviously, the infiltration was very slow compared with the layered movement of soil water in the dam farmland. It should be noted, however, also that the apparent velocity was not conclusive because no consideration for effects of evaporation or plant was made. Consequently, funnel flow, instability of wetting front, and water movement in layered soils were all found in the dam farmland.

4 Conclusions

The major conclusions of the present study in the dam farmland without tillage on the Loess Plateau of China are listed below:

- (1) The funnel flow was only found in the shallow soils with layered soil structure. This proved the existence of this phenomenon in the dam farmland as in the laboratory experiments and other field observations carried by other researchers.
- (2) The instability of an unsaturated wetting front as it passes from fine to coarse layers was demonstrated in the dam farmland. The existence of wetting front instability, carrying water rapidly and deeply in layered soils, might have great practical importance to the hydrologic process and environment protection.
- (3) The soil water moved laterally in the dam farmland proved by the spatial correlation analysis. However, the water movement mainly happened on the first day after rainfall with a velocity of about 7 m day^{-1} which was quicker than the infiltration rate. Not only by the preferential flow, but also the layered water flow should be considered in case of possible contamination in the dam farmland. More attention should be paid to the preferential flow and layered flow aiming at water resource management and contamination control.

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